

Material Safety Data Sheet

Product name

ZICOS PR-HE Zero

1. Product and company identification

- a) Product Name ZICOS PR-HE Zero
- b) Recommended use of the chemical and restrictions on use
- | | |
|---------------------------------|--|
| Recommended use of the chemical | Cosmetic additive(moisturizer,anti-microbia, l)Manufacture of substances |
| restrictions on use | Not available |
- c) Manufacturer/Supplier/Distributor Information
- | | |
|------------------------|---|
| Name | ZIKO |
| Address | 688 Chokumro, Kumwangeup, Eumsunggun, Chungbuk, Korea |
| Emergency phone number | 82-43-877-3300 |

2. Hazards identification

- a) Hazard Classification
- skin irritation (Category 2)
- Eye irritation (Category 2)
- Specific target organ toxicity – single exposure (Category 3), H335
- Xi Irritant R36/37/38

- b) Label elements including precautionary statements

Symbol



Single word

Warning

Hazard Statement

H315 Causes skin irritation.

H319 Causes serious eye irritation.

H335 May cause respiratory irritation.

Precautionary Statement
Precaution

P261 Avoid breathing dust/fume/gas/mist/vapours/spray.

P264 Wash skin thoroughly after handling.

P280 Wear eye protection/ face protection.

P302+P352 IF ON SKIN: Wash with plenty of soap and water.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes.
Remove contact lenses, if present and easy to do. Continue rinsing.

P337+P313 If eye irritation persists: Get medical advice/attention.

Storage

No available

Disposal

No available

- c) Other Hazard which are not included in the classification criteria (NFPA)

Health

2

Fire

1

Reactivity

0

3. Composition/Information on ingredients

Product name	ZICOS PR-HE Zero
IUPAC name	Hexane-1,2-diol (1,2-Hexanediol)
CAS No.	6920-22-5
Concentration(%)	99%

4. First aid measures

a) general advice	Consult a physician. Show this safety data sheet to the doctor in attendance. Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes. Do not use an eye ointment. Seek medical attention.
b) Eye contact	In case of contact, immediately flush skin with plenty of water.
c) Skin contact	Cover the irritated skin with an emollient.
d) Inhalation	If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention. Do NOT induce vomiting unless directed to do so by medical personnel.
e) Ingestion	Never give anything by mouth to an unconscious person. If large quantities of this material are swallowed, call a physician immediately. Loosen tight clothing such as a collar, tie, belt or waistband.

5. Fire-Fighting measures

a) Suitable (and unsuitable) extinguishing media	Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
b) Specific hazards arising from the chemical	Carbon oxides
c) Special protective equipment and precautions for fire-fighters	Wear self-contained breathing apparatus for firefighting if necessary.

6. Accidental release measures

a) Personal precautions, protective equipment and emergency procedures	Use personal protective equipment. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas.
b) Environmental precautions and protective procedures	Do not let product enter drains.
c) Methods and materials for containment and cleaning up	Soak up with inert absorbent material and dispose of as hazardous waste. Keep in suitable, closed containers for disposal.

7. Handling and storage

a) Precautions for safe handling	Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Normal measures for preventive fire protection.
b) Conditions for safe storage (including any incompatibilities)	Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

8. Exposure controls & personal protection

a) Control parameters (e.g. occupational exposure limit values, biological limit values)

Domestic provision

Not available

ACGIH provisions

Not available

Biological Exposure standard

Not available

b) Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

c) Personal protective equipment

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU). Safety glasses with side-shields conforming to EN166 Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands. The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it. impervious clothing, The type of protective equipment must be selected according to the

Eye protection

Skin protection

Body protection

concentration and amount of the dangerous substance at the specific workplace.

9. Physical and chemical properties

a) Appearance	
physical state	Liquid
Color	colourless
b) Odor	Not available
c) Odor threshold	Not available
d) pH	Not available
e) Meting point/freezing point	2℃
f) Initial boiling point and boiling range	228~229℃
g) Flashing point	122 ℃ – closed cup
h) Evaporation rate	Not available
i) Flammability (solid, gas)	Not available
j) Upper/lower flammability or explosive limits	Not available
k) Vapor pressure	0.576 Pa (at 25℃)
l) Solubility	>9g/g (in water)
m) Vapor density	Not available
n) Relative density	0.951g/cm3 at 25℃
o) Partition coefficient: n-octanol/water	0.58
p) Auto-ignition temperature	350℃
q) Decomposition temperature	Not available
r) Viscosity	87.8mPas(kinematic viscosity) at 20℃ / 27.7mPas(kinematic viscosity) at 40℃
s) Formula mass	118.17

10. Stability and reactivity

a) Chemical stability and possibility of hazardous reactions	Stable under recommended storage conditions.
b) Conditions to avoid	Avoid excessive heat and light.
c) Incompatible materials	Strong oxidizing agents, Strong alkalis (bases)
d) Hazardous decomposition products	In the event of fire : Carbon oxides

11. Toxicological information

a) Information on the likely routes of exposure

Not available

b) Health hazards information

Acute toxic

oral

Not available

dermal

Not available

inhalation

Not available

Skin corrosive/irritant

Not available

Serious eye damage/eye irritation

Not available

Respiratory sensitization

Not available

Skin sensitization

Not available

Carcinogenicity

IARC

No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

OSHA

Not available

ACGIH

Not available

NTP

Not available

EU CLP

Not available

Germ Cell Mutagenicity

Not available

Reproductive toxicity

Not available

Specific target organ toxicity
(single exposure)

Inhalation – May cause respiratory irritation.

Specific target organ toxicity
(repeated exposure)

Not available

Aspiration hazard

Not available

12. Ecological information

a) Aquatic and terrestrial ecotoxicity

Fish

LC50 1291.099 mg/l 96 hr (estimate)

Crustacea

LC50 599.986 mg/l 48 hr (estimate)

Algae

EC50 161.332 mg/l 96 hr (estimate)

b) Persistence and degradability

Remaining characteristic

log Kow 0.58

Resolvability

Not available

c) Bioaccumulative potential

Biodegradation

BCF 3.162 (estimate)

Bioconcentration factor

Not available

d) Mobility in soil

Not available

e) Result of PBT vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted

f) Other adverse effects

Not available

13. Disposal considerations

a) Disposal method

Product

Offer surplus and non-recyclable solutions to a licensed disposal company.
Contact a licensed
professional waste disposal service to dispose of this material.

b) Disposal precaution

Dispose of contents / container in accordance with local regulation.

14. Transport information

- | | |
|--|---|
| a) UN number(UN No.) | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| b) UN proper shipping name | ARD/RID : Not dangerous goods
IMDG : Not dangerous goods
IATA : Not dangerous goods |
| c) Transport hazard class | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| d) Packing group (if applicable) | ARD/RID : Not applicable
IMDG : Not applicable
IATA : Not applicable |
| e) Environmental hazards | ARD/RID : Not applicable
IMDG Marine pollutant : Not applicable
IATA : Not applicable |
| f) Special precaution which a user to be aware of or needs to comply with in connection with transport or conveyance either within or outside their premises | |
| Type of fire emergency action | Not applicable |
| Type of exposure emergency actions | Not applicable |

15. Regulatory information

- | | |
|--|--|
| a) Industrial Safety and Health Act | Not applicable |
| b) Toxic Chemical Control Act | Not applicable |
| c) Dangerous Material Safety Control Act | Not applicable |
| d) Wastes Management Act | Not applicable |
| e) Other regulations | |
| Federal and State Regulations | TSCA 8(b) inventory: DL-1,2-Hexanediol |
| EU Regulations(EINECS) | This product is on the European Inventory of Existing Commercial Chemical Substances. |
| f) Other classifications | |
| WHIMS(Canada) | Not controlled under WHMIS (Canada). |
| DSCL (EEC) | R36/38- Irritating to eyes and skin.
S2- Keep out of the reach of children.
S46- If swallowed, seek medical advice immediately and show this container or label. |
| HMIS (U.S.A.) | Health Hazard (2) / Fire Hazard (1) / Reactivity (0) / Personal Protection (h)
National Fire Protection Association (U.S.A.):
Health (1) / Flammability (1) / Reactivity (0) / Specific hazard: none /
Protective Equipment
(Gloves, Lab coat) |

16. Other information

a) Information source and references

ECHA(a) appearance)
ECHA(e) Melting point/freezing point)
ECHA(f) Initial boiling point and boiling range)
ECHA(g) Flashing point)
ECHA(k) Vapor pressure)
ECHA(l) Solubility)
ECHA(o) Partition coefficient: n-octanol/water)
ECHA(p) Auto-ignition temperature)
ECHA(r) Viscosity)
ECHA:(Skin corrosive/irritant)
ECHA(Serious eye damage/eye irritation)
EPI SUITE(Fish)
EPI SUITE(Crustacea)
EPI SUITE(Algae)
ECHA(Remaining characteristic)
EPI SUITE(Biodegradation)
EPI SUITE(Mobility in soil)

b) Issuing date 2017-03-23

c) Revision number and date

Revision number 0
Revision date 0

d) others

- The above information is believed to be correct but does not claim to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. Employers should use this information only as a supplement to other information gathered by them, and should make independent judgement of suitability of this information to ensure proper use and protect the health and safety of employees. It does not represent any guarantee of the properties of the product. ZIKO CO., LTD. shall not be held liable for any damage resulting from handling or from contact with the above product.